Manipulation Planning

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Where this fits:

- **Builds on:** Symbolic task planning; Configuration space; Minkowski sum; Lazy collision checking; Narrow passages and smarter sampling; Probabilistic roadmap (PRM).

1. Overview

Key point: When a robot moves an object, the object is an obstacle while it stands and part of the robot while it is held, so the free space changes with each grasp and release. Manipulation planning splits the task into transit and transfer motions joined where both are possible; slow pushes move objects that cannot be held, and planning backwards from the goal with motions that slide along surfaces makes the result certain despite errors in the hand's position.

So far the robot plans its own motion among obstacles that never move. The probabilistic roadmap (G-2780) answers many trips in an unchanged room, and symbolic task planning (G-2600) puts the steps of carrying a cup in order: go to the table, pick the cup up, carry it, put it down. "Pick up" is one step to the task planner, but the robot still has to move its arm to do it.

Manipulation needs an arm, so this Note leaves the room's floor for the table top. We use the arm whose configuration (G-2286), the numbers that place it, is its two joint angles: two links, 0.3 m and 0.2 m long, each turning about its joint, with the robot parked beside the table and the arm seen from the side. The shoulder is 0.1 m above the table top, and a suction gripper at the hand holds a cup by its lid while a wrist keeps the cup upright.

The task: move the cup, 6 cm wide and 8 cm tall, from spot A on the table to spot B, 0.24 m further on, with a bottle 0.14 m tall standing between them. The difficulty is the cup itself, which is neither an obstacle that never moves nor a part of the robot:

- planned as a fixed obstacle, the cup can never move;
- planned as if it were not there, the cup goes wherever the hand goes. Figure 1 follows a route planned for the arm alone, 109.3 degrees of joint motion on which the arm touches nothing (notebook): the cup it carries sinks 1.7 cm into the table and passes through the bottle for 32.2 percent of the way.

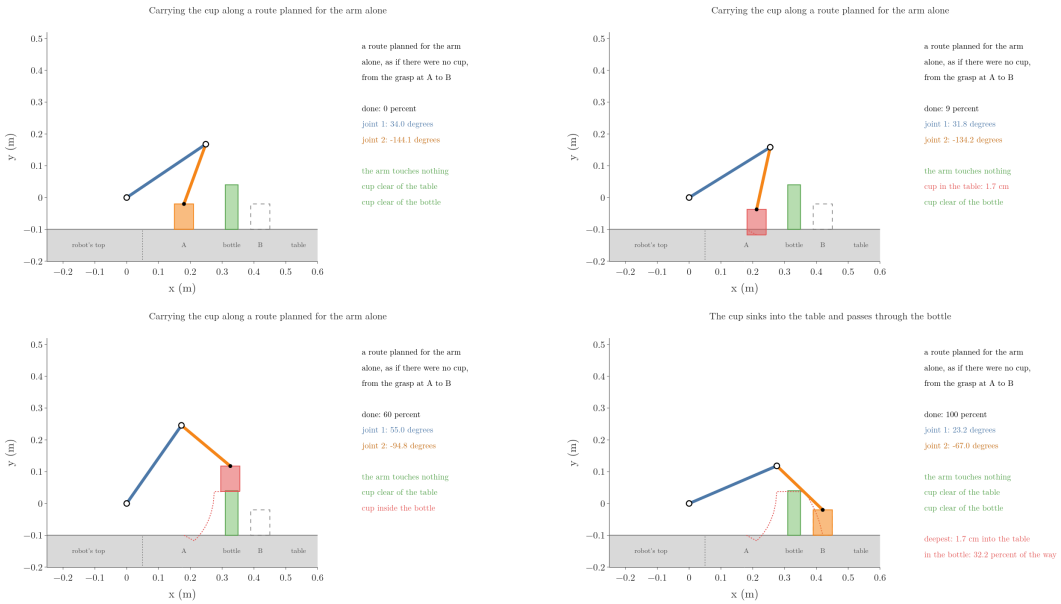


Figure 1: Animation: the arm (blue link 1, orange link 2) carries the cup from spot A along a route planned for the arm alone, as if there were no cup (a probabilistic roadmap in the arm's configuration space, seed 0, 109.3 degrees of joint motion). The robot's top and the table are grey, the bottle green, spot B dashed. The arm touches nothing; the cup, drawn red while it overlaps something, sinks up to 1.7 cm into the table early on and passes through the bottle for 32.2 percent of the way; the dotted red line traces the cup's bottom. Last frame: the cup at B

Two more problems remain. A cup without a lid cannot be held by suction at all, yet it still has to move. And the hand is never exactly where it is told to go: if its position is known only to within 1 cm, a cup holder at B that leaves 0.5 cm of play on each side will often be missed.

The opening question of this Note: **how does the robot plan motions that move objects, by holding them or by pushing them, and still end with the cup where it must be?** We answer it in steps:

- carrying the cup: motions with and without the cup, and where they meet (Section 2);
- moving the cup without holding it, by a slow push (Section 3);
- placing the cup despite the hand's error, by planning backwards from the goal (Section 4);
- the software real robots use for all of this (Section 5).

2. Carrying the cup: transit and transfer

> **Key point:** The task has two kinds of motion: transit, with the hand empty and the cup an obstacle at its spot, and transfer, with the cup held and moving with the hand. Each has its own free space; the mode changes only at a configuration free in both, the grasp or the placing. Planning each motion in its own free space, with the PRM, moves the cup from A to B.

2.1 Two kinds of motion, two free spaces

The opening problem was that the cup is sometimes an obstacle and sometimes part of the robot. Which one depends on whether the hand holds it, so we split the task where that changes:

- while the hand is empty, the arm moves and the cup stays where it stands, so the cup is an obstacle like the bottle;
- while the hand holds the cup, the cup moves with the hand as if it were part of the arm, so it must not hit the table, the bottle or the arm's own links.

A motion of the first kind is a **transit** (G-2801): the robot moves while the object stays where it is. A motion of the second kind is a **transfer** (G-2802): the object is held and moves as if attached to the robot's hand (LaValle §7.3.2; Hauser ch.12, "Multi-modal motion planning"). Which of the two applies, together with where the object stands or how it is held, is the **mode** (G-2803) of the task (Hauser ch.12).

Each mode has its own free space. The arm's configuration is its two joint angles, so its C-space is a square of angles from $-180°$ to $180°$, whose opposite edges are the same angles (cutting the torus into a square). Figure 2 draws the square three times, once per stage of the task, and colours every blocked configuration by what it collides with: grey the table or the robot's top, green the bottle, orange the cup in the way. Watch the orange part move: round the grasp in the first panel, where the links would hit the standing cup; at the top and bottom edges in the second, where the held cup would hit link 1; round the placing in the third.

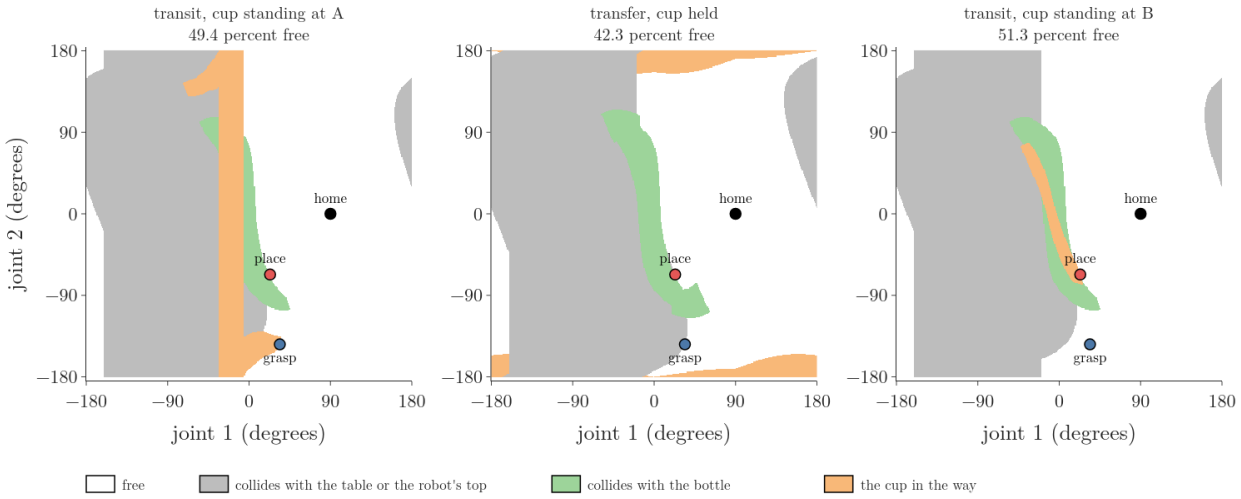


Figure 2: The arm's C-space, joint 1 across and joint 2 up, each from −180 to 180 degrees, in the three stages of the task; each blocked configuration is coloured by what it collides with: grey the table or the robot's top, green the bottle, orange the cup in the way. Left, transit with the cup standing at A: 49.4 percent free. Middle, transfer with the cup held under the hand: 42.3 percent free; the held cup closes extra bands at the top and bottom and widens the bottle's region. Right, transit with the cup standing at B: 51.3 percent free. Black: home (90, 0) degrees; blue: the grasp at A (34.0, −144.1); red: the placing at B (23.2, −67.0)

The free share changes from 49.4 percent to 42.3 percent and then to 51.3 percent (notebook). A single free space for the whole task cannot exist, because the cup that blocks the arm in transit is part of the arm in transfer. Section 2.2 asks where one mode can hand over to the other.

2.2 Where the modes meet: transition configurations

A transit can end and a transfer begin only where both are possible: the hand on the cup's lid, ready to lift, touching the cup but not overlapping it (LaValle §7.3.2). The joint angles that put the hand on the lid follow from the triangle of the two links and the line from the shoulder to the lid. Of its two solutions, elbow up and elbow down, only the elbow-up one is free (notebook):

- the grasp at A: joint 1 at 34.0°, joint 2 at −144.1°, hand on the lid at (0.18, −0.02) m from the shoulder;
- the placing at B: joint 1 at 23.2°, joint 2 at −67.0°, hand at (0.42, −0.02) m.

The grasp is free in the transit mode, because link 2 only touches the lid, and free in the transfer mode, because the held cup rests on the table without sinking in (notebook). A configuration that is free in two modes, where the task can switch from one to the other, is a **transition configuration** (G-2804) (Hauser ch.12; LaValle §7.3.2). In Figure 2, the blue grasp sits on the edge of the orange region in the first panel: one step further and link 2

would push into the cup.

The arm's position at each moment follows from its two joint angles by adding the two links end to end, as the arm's configuration (G-2286) showed with the hand at (0.260, 0.350) m for the angles (30°, 60°). With the transition configurations known, the three motions between them can be planned.

2.3 The plan: transit, transfer, transit

The plan has three legs, each in one mode, joined at the transition configurations (Hauser ch.12, "Simple Pick-and-Place"):

1. transit, with the cup standing at A, from home to the grasp;
2. grasp, then transfer, with the cup held, from the grasp to the placing;
3. release, then transit, with the cup standing at B, from the placing back home.

Each leg is an ordinary planning problem in its own free space, so the probabilistic roadmap (G-2780) plans it: 150 milestones (G-2781), random free configurations, each joined to its 10 nearest, with distances measured the short way round each joint (Hauser ch.12: "using your favorite C-space planner"). Figure 3 runs the plan. Watch the C-space on the right change at the two mode switches, and the path stay in the white free space of whichever mode is current.

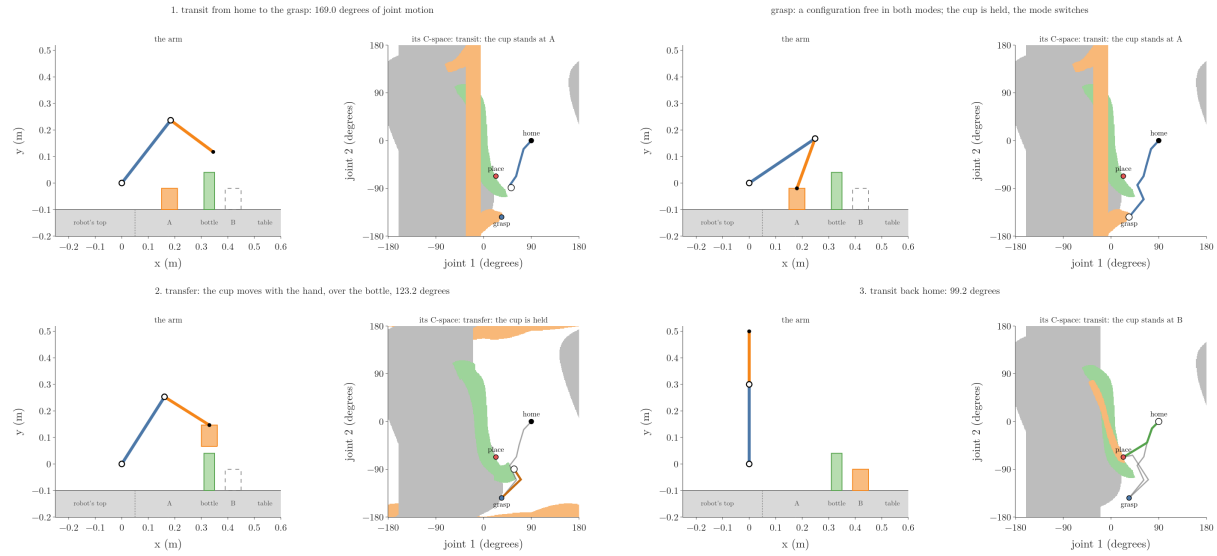


Figure 3: Animation: the plan, transit, transfer, transit. Left: the arm, with the cup standing at A, then held under the hand, then standing at B. Right: the arm's C-space of the current mode (coloured as in Figure 2) with the path so far. Leg 1, transit from home to the grasp, 169.0 degrees of joint motion (blue). At the grasp the cup is held and the C-space switches to the transfer mode. Leg 2, transfer over the bottle to the placing, 123.2 degrees (orange). At the placing the cup is let go and the C-space switches back. Leg 3, transit home, 99.2 degrees (green); earlier legs stay grey

The three roadmaps give routes of 169.0, 123.2 and 99.2 degrees of joint motion, all free in their modes (notebook). During the transfer, the bottom of the held cup rises to 0.076 m, 0.036 m above the bottle's top at 0.04 m:

$$0.076 - 0.04 = 0.036 \text{ m}$$

So the cup clears the bottle that the route of Figure 1 dragged it through: the transfer was planned in the free space where the cup is part of the robot.

The formal version. The state of the task is the arm's configuration together with the cup's position, and the mode says which of the two free spaces applies (Hauser ch.12). A plan is a sequence of transit and transfer paths that alternate, each inside its mode's free space, switching only at transition configurations (LaValle §7.3.2). Moving a robot and the objects it handles from where they are to where they should be is **manipulation planning** (G-2800) (LaValle §7.3.2). The symbolic plan's "pick up" is now a transit that ends at a transition configuration, then a switch of mode. Every leg so far needed the cup to be held, though, and a cup without a lid cannot be.

3. Moving without grasping: pushing and the quasistatic assumption

> **Key point:** An object that cannot be held can still be pushed. If the push is slow, speeding the object up takes a negligible force, so the forces on it balance at every moment: this quasistatic assumption makes the object move only while it is pushed, so where the hand stops decides where the object stops. Pushing the cup low makes it slide; pushing it more than 7.5 cm above the table tips it over.

3.1 Manipulation without a grasp

The opening's second problem: a cup without a lid. A suction gripper holds an object only by a flat surface that seals (Hauser ch.12, "Suction grippers"), so the open cup cannot be held. The robot can still move it by touching it: pushing it along the table with the hand. Manipulating an object without grasping it, by pushing, tilting, throwing or squeezing it, is **nonprehensile manipulation** (G-2805) (LaValle §12.5.2; MR §12.3, which adds pushing a sofa and carrying glasses on a tray). Nonprehensile manipulation also handles objects too large to grasp or too heavy to lift (MR §12.3).

The task for this section: push the open cup 6 cm towards the robot, from A (centre at x = 0.18 m) to a spot C (x = 0.12 m). The plan is the hand's motion: touch the cup's right side, move 6 cm to the left, stop. For that to work, the cup must stop when the hand stops.

3.2 Why a slow push is predictable

Figure 4 pushes the cup three ways. In part 1, watch the cup slide with the hand at 0.05 m/s and stop with it at C; in part 3, watch it slide on past C after a push at 0.5 m/s.

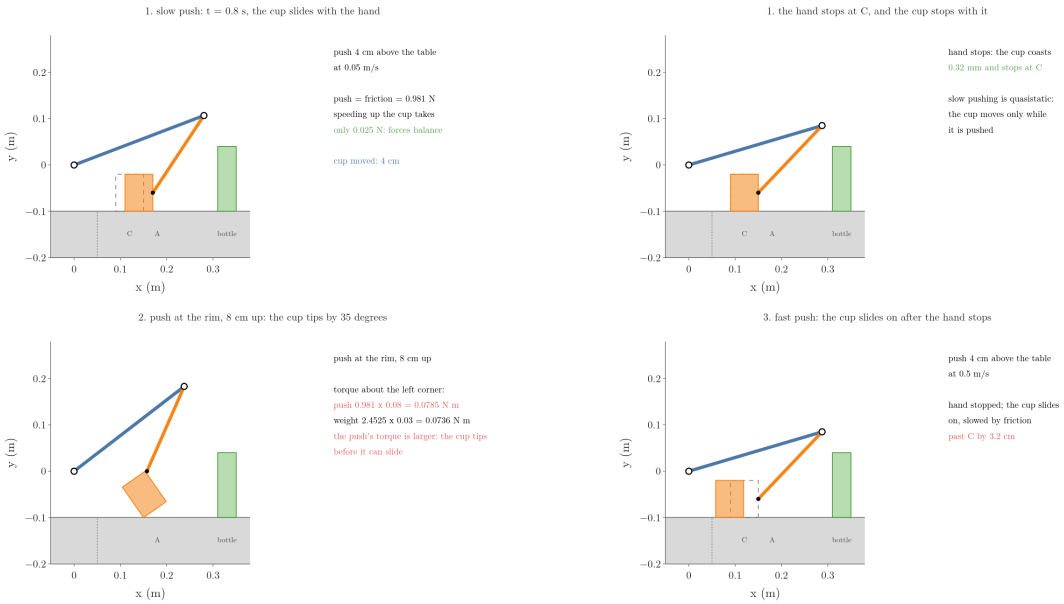


Figure 4: Animation: the arm pushes the open cup (no lid) to the left, its hand touching the cup's right side; the target spot C is dashed. Part 1, a slow push at 0.05 m/s, 4 cm above the table: the cup slides with the hand, the push equals the friction, 0.981 N, and speeding the cup up takes only 0.025 N; when the hand stops at C, the cup coasts 0.32 mm and stops at C. Part 2, a push at the rim, 8 cm up: the push's torque about the cup's left bottom corner, $0.981 \times 0.08 = 0.0785$ N m, exceeds the weight's, $2.4525 \times 0.03 = 0.0736$ N m, so the cup tips over that corner and falls. Part 3, a fast push at 0.5 m/s, 4 cm up: the hand stops at C, and the cup slides on, slowed by friction, 3.2 cm past C

The cup's numbers (the coefficient of friction is an assumed value for a cup on a table):

- m is the cup's mass, 0.25 kg;
- g is the acceleration due to gravity, 9.81 m/s^2;
- μ is the coefficient of friction between the cup and the table, 0.4, a ratio with no unit.

The cup presses on the table with its weight:

$$mg = 0.25 \times 9.81 = 2.4525 \text{ N}$$

The table's friction can hold back at most μ times the force pressing the two surfaces together (MR §12.2, Coulomb friction); that ratio is the **coefficient of friction** (G-2807). With 0.4, every newton that presses the cup onto the table lets friction hold back at most 0.4 N. So the friction on the sliding cup is:

$$\mu mg = 0.4 \times 2.4525 = 0.981 \text{ N}$$

Why the forces balance. The force left over after friction speeds the cup up, by **Newton's second law** (G-2809): the net force on a body, all the forces on it added with their directions,

equals its mass times its acceleration, how fast its speed changes (MR §8.1.1, eq. 8.4). With:

- F_{net} is the net force on the cup, the push minus the friction, in N;
- a is the cup's acceleration, how fast its speed grows, in m/s^2.

$$F_{\text{net}} = ma$$

A slow push brings the cup from rest to 0.05 m/s in about 0.5 s:

$$a = 0.05/0.5 = 0.1 \text{ m/s}^2$$

$$F_{\text{net}} = 0.25 \times 0.1 = 0.025 \text{ N}$$

$$\text{share} = 0.025/0.981 = 0.0255$$

So speeding the cup up takes 2.5 percent of the friction force: the push almost exactly balances the friction at every moment. When the speeds and accelerations are so small that the force needed to change the speed can be ignored, the forces on the object balance at every instant; that is the **quasistatic assumption** (G-2806) (MR §12.3, eq. 12.32).

Why the cup stops with the hand. When the hand stops, only friction acts on the cup, and it slows the cup down. With:

- v is the cup's speed when the hand stops, in m/s;
- b is how fast friction slows it, in m/s^2;
- t is the time it takes to stop, in s;
- d is the distance it slides on, in m.

By Newton's second law, friction alone slows the cup by the friction force divided by the mass. After the fast push of part 3, the cup starts at 0.5 m/s:

$$b = 0.981/0.25 = 3.924 \text{ m/s}^2$$

$$t = 0.5/3.924 = 0.1274 \text{ s}$$

The speed falls steadily from 0.5 m/s to 0, so its average is half the start speed:

$$d = (0.5/2) \times 0.1274$$

$$= 0.0319 \text{ m}$$

The same steps in symbols:

$$b = \mu m g / m = \mu g$$

$$t = v / b$$

$$d = (v/2)\, t$$

$$= \frac{v^2}{2b}$$

At 0.05 m/s, ten times slower, the distance is a hundred times smaller:

$$d = \frac{0.05^2}{2 \times 3.924} = 0.00032 \text{ m}$$

So the slow push of part 1 leaves the cup within 0.32 mm of where the hand stopped, while the fast push of part 3 overshoots C by 3.2 cm (notebook). Under the quasistatic assumption the cup moves only while it is pushed, so the plan only has to say where the hand goes, not how the cup speeds up and slows down (MR §12.3). The push's height, though, decides whether the cup slides at all.

3.3 Where to push: slide or tip

Part 2 of Figure 4 pushes at the rim and the cup tips over instead of sliding. The cup can do one of three things under a push: remain at rest, slide, or tip over its front bottom corner, the left one when pushed to the left. Quasistatic analysis tests each for balance and keeps the one that balances (MR §12.3).

The turning effect of a force about a point is its **torque** (G-2808): the force times its distance from the point, measured square to the force's line (MR §3.4). A door shows why the distance matters: the same push turns it more easily at the handle, far from the hinges, than close to them. Figure 5 marks the forces on the cup and their distances from the corner O.

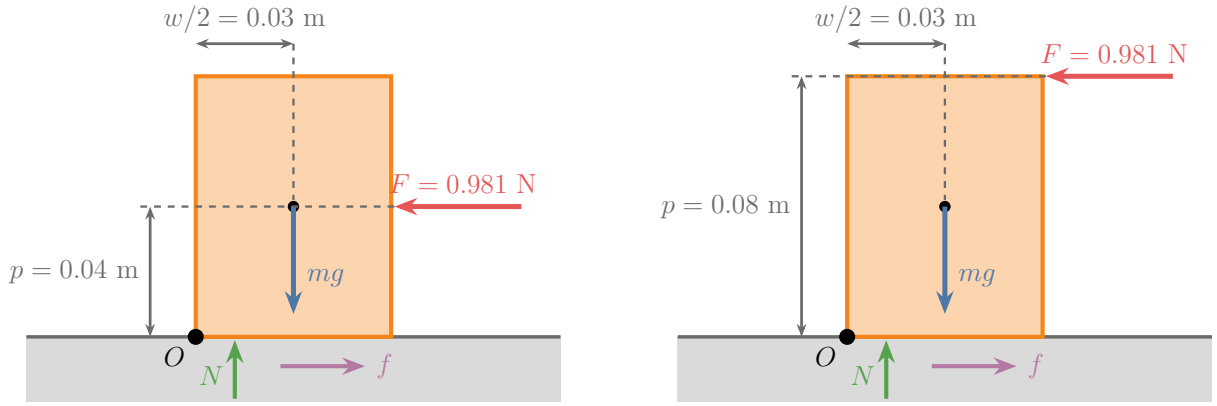


Figure 5: The forces on the pushed cup, side view, drawn 50 times its size: the push F = 0.981 N to the left at height p, the weight mg = 2.45 N down through the centre, the table's support N and the friction f along the base; O is the left bottom corner. Left, p = 0.04 m: the push's torque about O is 0.981 × 0.04 = 0.0392 N m, less than the weight's 2.4525 × 0.03 = 0.0736 N m, so the cup slides. Right, p = 0.08 m: 0.981 × 0.08 = 0.0785 N m, more than 0.0736 N m, so the cup tips

The support and the friction act at the base, through O or along a line through it, so their torques about O are zero. The weight acts half the cup's width from O and holds the cup down; the push acts at height p and turns it over. To slide, the push must reach the friction force, 0.981 N. Pushing 4 cm up:

$$\text{push: } 0.981 \times 0.04 = 0.0392 \text{ N m}$$

$$\text{weight: } 2.4525 \times 0.03 = 0.0736 \text{ N m}$$

The weight's torque is larger, so the cup slides flat. Pushing at the rim, 8 cm up:

$$\text{push: } 0.981 \times 0.08 = 0.0785 \text{ N m}$$

The push's torque exceeds the weight's before the push is large enough to slide the cup, so the cup tips. The same steps in symbols, with:

- μmg is the friction force, 0.981 N, and mg the cup's weight, 2.4525 N (Section 3.2);
- p is the push's height above the table, in m;
- w is the cup's width, 0.06 m, so its centre is $w/2$ from O.

The cup slides without tipping if the push's torque at the sliding force stays below the weight's:

$$\mu m g\, p \le m g\, w/2$$

$$p \le \frac{w}{2\mu}$$

$$= 0.06/(2 \times 0.4) = 0.075 \text{ m}$$

A push below 7.5 cm slides the cup; a push above it tips the cup (MR §12.3, Exercise 12.23 asks the same of a pushed box). The robot's plan for the open cup is now complete: touch the cup's side 4 cm up, move the hand 6 cm to the left slowly, stop. A **nonprehensile** (G-2805) plan like this needs no grasp and, under the quasistatic assumption, no sensing while it runs; LaValle §12.5.2 shows even sequences of squeezes that turn a part to a known orientation without any sensor. Every plan so far, though, assumed that the hand goes exactly where it is told.

Another way to see it. A metre stick rests on two index fingers, one finger near the centre and one far from it, and the fingers slowly slide together. The stick never falls, and the fingers meet under its centre (MR §12.3, Example 12.10, the meter-stick trick). Testing each way the stick could slip for force balance, one at a time, shows why:

- the finger farther from the centre carries less of the stick's weight, so less friction, and the stick slides over it while it rides along on the other finger;
- once the centre is halfway between the fingers, the stick slips on both, and the centre stays between them until they meet.

The analysis holds only for slow motion: if a finger moves fast, the stick falls, because the quasistatic assumption no longer holds (MR §12.3).

4. Placing despite errors: preimage planning

> **Key point:** A motion command whose direction can be off by up to 10 degrees surely reaches the goal only from a set of starts, its preimage. Sliding along the holder's chamfers turns the preimage from a sliver 1 cm wide into a triangle 3 cm wide and 8.5 cm tall, so a hand known only to within 1 cm still places the cup. Planning backwards, each command must end inside the preimage of the next.

4.1 The error that makes the placing fail

The opening's third problem: the hand is never exactly where it is told. Suppose the robot knows the hand's position only to within 1 cm either side, and any straight motion it commands may run up to 10 degrees off its direction (an error model of the kind preimage planning assumes, LaValle §12.5.1). B is now a cup holder, so that the cup cannot be knocked

over (Figure 6, left): two posts 2 cm tall, 7 cm apart at the bottom, each with a 45-degree chamfer 1 cm wide on its inner top corner. The 6 cm cup has 0.5 cm of play on each side.

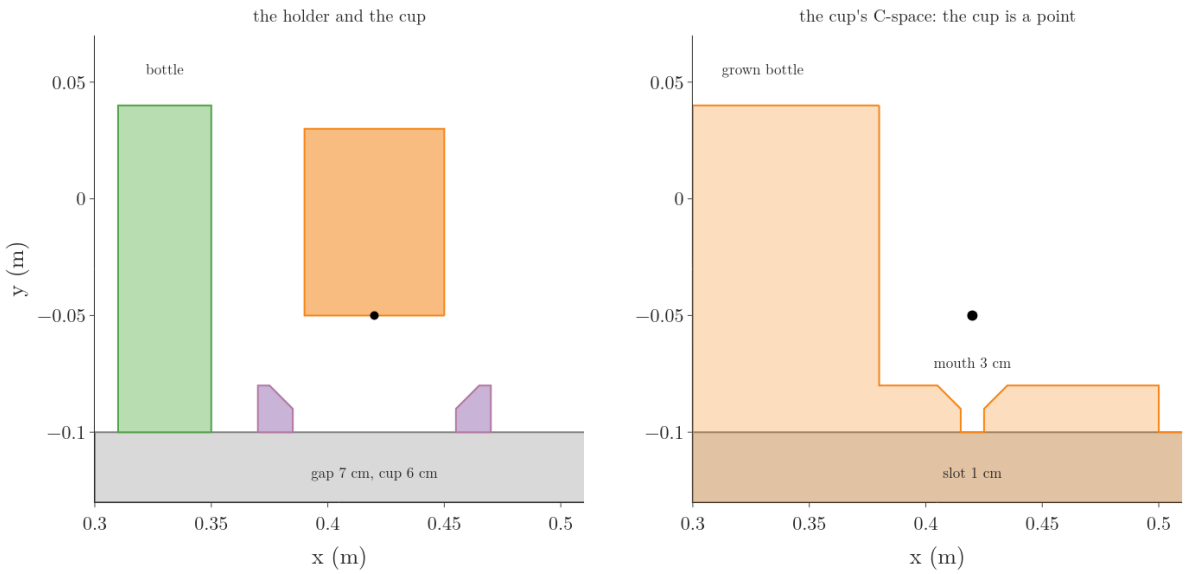


Figure 6: The cup holder at B. Left, side view: the bottle, the two posts of the holder (purple) 1.5 cm thick and 2 cm tall, 7 cm apart at the bottom, each with a 45-degree chamfer, and the 6 cm cup above them; the black dot is the middle of the cup's bottom. Right, the cup's C-space: the cup becomes that point, and the posts, the bottle and the table grow by the cup's shape (orange); the holder becomes a slot 1 cm wide at the bottom whose chamfers open to a mouth 3 cm wide, 2 cm above the table

To reason about the cup as a point, we use its C-space. The cup only moves, never turns, so as for the board of the Minkowski sum (G-2653), each obstacle grows by the cup's shape reflected, and the cup becomes the point at the middle of its bottom (Figure 6, right). The gap between the posts shrinks by the cup's width:

$$0.07 - 0.06 = 0.01 \text{ m}$$

So the slot is 1 cm wide, and the chamfers widen its mouth by 1 cm on each side, to 3 cm. A hand within 1 cm either side of the slot's centre spans 2 cm, twice the slot, so aiming straight at the slot cannot be guaranteed to work.

4.2 The preimage of the holder

The command is "move down until stopped". From a given start, every direction within 10 degrees of straight down is possible, so the start is good only if every one of them ends in the holder. Figure 7 draws five of those directions from one start, 0.8 cm right of the slot's centre and 2 cm above the posts. In the stage "without sliding", watch four of the five land

on the chamfer and stop; in the stage “with sliding”, watch the same four slide down the chamfer into the slot.

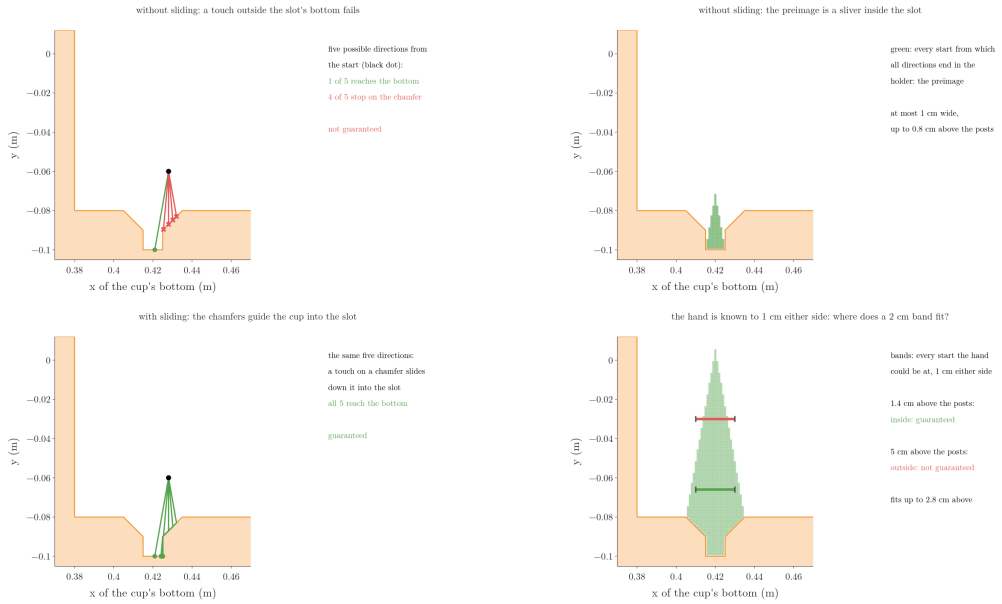


Figure 7: Animation in the cup’s C-space near B (orange: the posts, bottle and table grown by the cup’s shape; the slot is 1 cm wide). The command “move down” may run up to 10 degrees off; five possible directions (260 to 280 degrees) are drawn from a start (black dot) 0.8 cm right of the slot’s centre and 2 cm above the posts. Without sliding, a touch outside the slot’s bottom fails: 1 of the 5 reaches the bottom, 4 stop on the chamfer (red). The preimage without sliding, the starts from which every direction ends in the holder (green), is a sliver at most 1 cm wide reaching 0.8 cm above the posts. With sliding, the chamfers guide all 5 into the slot. The preimage with sliding is a triangle 3 cm wide at the posts’ top and 8.5 cm tall. Last frame: a band 2 cm wide, every start the hand could be at, fits inside it 1.4 cm above the posts (green) but lies outside it 5 cm above them (red)

The set of starts from which a motion command is sure to reach its goal, whatever error occurs, is the **preimage** (G-2811) of the goal for that command (LaValle §12.5.1). Without sliding, the preimage is a sliver inside the slot, at most 1 cm wide and reaching only 0.8 cm above the posts (notebook): the robot would already have to be in the slot.

Sliding is what helps. On touching a surface, a moving object either sticks or slides along it. A motion that slides along a surface while touching it is a **compliant motion** (G-2810) (LaValle §12.5.1). Friction stops the sliding when the motion points nearly straight into the surface: with the coefficient of friction 0.4, sliding needs the motion to point more than 21.8 degrees away from straight into the surface, the angle whose tangent is 0.4, because the part of the motion along the surface must exceed 0.4 times the part pressing into it (LaValle §12.5.1; MR §12.2):

- on a chamfer, tilted 45 degrees, a motion within 10 degrees of straight down points 35 to 55 degrees away from straight into it, so the cup slides down the chamfer into the

slot;
- on the flat top of a post, a motion within 10 degrees of straight down points at most 10 degrees away from straight into it, so the cup sticks there and the placing fails.

With sliding, the preimage is a triangle 3 cm wide at the posts' top and 8.5 cm tall (notebook). The flat tops of the posts, where the cup would stick, bound it on both sides.

4.3 How high the hand may start

A drop of height h drifts sideways by at most h times the tangent of 10 degrees. With:

- h is the height of the start above the posts' top, in m;
- 0.176 is the tangent of 10 degrees;
- 0.01 m is the hand's error either side;
- 0.015 m is half the mouth's width.

Every start in the hand's 2 cm band lands inside the mouth, and slides into the slot, if the error plus the drift stays within half the mouth. For a start 2 cm up:

$$0.01 + 0.02 \times 0.176 = 0.0135 \text{ m}$$

$$0.0135 \leq 0.015$$

so a 2 cm drop is safe. The highest safe start:

$$0.01 + 0.176\,h \leq 0.015$$

$$h \leq 0.005/0.176 = 0.028 \text{ m}$$

The same steps in symbols, with:

- e is the hand's error either side, 0.01 m;
- s is half the mouth's width, 0.015 m;
- α is the largest error in the motion's direction, 10 degrees.

$$e + h \tan\alpha \leq s$$

$$h \leq \frac{s - e}{\tan\alpha}$$

So the hand may start the drop anywhere up to 2.8 cm above the posts (notebook: 0.0284 m), as the green band of Figure 7, 1.4 cm up, shows. Without sliding, every start must land in the 1 cm slot itself:

$$0.01 + 0.176\,h \leq 0.005$$

which no height h meets, because the hand's error alone, 0.01 m, is already more than 0.005 m.

4.4 Planning backwards from the goal

The drop works only if the motion before it ends inside the drop's preimage, and the preimage fixes the goal of the previous motion: the transfer of Section 2.3, the motion with the cup held, must now end with the cup 1.4 cm above the posts, centred over the slot, not at a placing configuration that assumes a perfect hand. Choosing each command so that it ends inside the preimage of the next, starting from the goal and working backwards, is **preimage planning** (G-2812) (LaValle §12.5.1, also called the LMT framework after Lozano-Pérez, Mason and Taylor). For the cup:

1. last command: move down until stopped, from the band 1.4 cm above the posts; its preimage is the triangle of Figure 7;
2. the command before: the transfer, planned with the roadmap, ending inside that band, so the hand's 1 cm error is covered;
3. before that: the transit to the grasp.

Real pick-and-place software does the same with short fixed motions: it plans to a configuration just above the target and finishes with a straight move down a centimetre or two (Hauser ch.12, "Pre-grasp, post-grasp, pre-placement, and post-placement motions"). Hauser gives a second reason: the last centimetres before a grasp or a placing are a deep narrow passage (G-2793), a thin part of free space, for a sampling planner, and a fixed straight motion avoids sampling it. Computing preimages for general problems is hard (LaValle §12.5.1), so such fixed final motions are planned by hand, as here. The robot can now carry, push and place the cup; the last section shows the software that does this on real robots.

5. The software: MoveIt and OMPL

> **Key point:** OMPL is a library of sampling-based planners, such as PRM and RRT, that by design contains no model of robots or collisions. MoveIt is the robot-arm software around it: it keeps a model of the robot and its surroundings, checks every configuration the planner tries (including an object attached to the hand, which is the transfer mode), and turns the planner's path into a timed trajectory.

The cup task of Sections 2 to 4 used a roadmap planner, a collision checker that accounts for whether the cup is held, and a model of the arm. Real robots get these pieces from two open-source packages.

OMPL (G-2814), the Open Motion Planning Library, is an open-source library (BSD licence, OMPL licence page) of sampling-based planners, among them PRM, RRT, EST and KPIECE, which work on abstractly defined spaces of configurations (OMPL). OMPL

contains no collision checker and no drawing, on purpose, so that it can be joined to any of them (OMPL). Its PRM is the roadmap of Section 2 of the probabilistic roadmaps.

MoveIt (G-2813) is motion-planning software for robot arms, free under the BSD licence (MoveIt home page); it uses OMPL's planners as its default planners, through a plugin interface that also accepts others (MoveIt docs, "Motion Planning"). MoveIt keeps the **planning scene** (G-2815): its model of the world round the robot and of the robot's own state, including any object attached to, that is carried by, the robot (MoveIt docs, "The move_group node"). Figure 8 follows one request, the transfer of the cup, through MoveIt. Watch the planner and the planning scene take turns: the planner proposes configurations, the scene answers free or not.

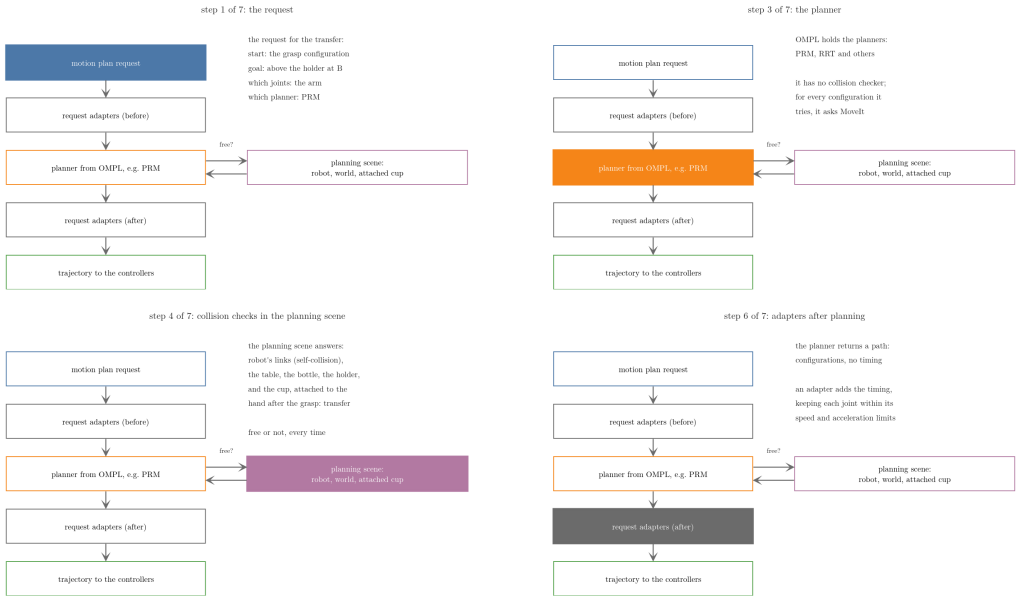


Figure 8: Animation: one motion plan request followed through MoveIt, one box highlighted per frame. 1, the request: start at the grasp configuration, goal above the holder at B, the arm's joints, the PRM planner. 2, request adapters before planning fix small problems, such as a start slightly outside a joint limit. 3, a planner from OMPL, here PRM; OMPL has no collision checker, so for every configuration it tries it asks MoveIt. 4, the planning scene answers: it checks the robot's links against each other, against the table, the bottle and the holder, and the cup attached to the hand after the grasp. 5, back to the planner. 6, request adapters after planning add timing, keeping each joint within its speed and acceleration limits. 7, the trajectory goes to the arm's controllers

Each piece of the cup task has its place in that pipeline (MoveIt docs, "Motion Planning"):

- **Transfer mode:** after the grasp, the cup is added to the planning scene as an object attached to the hand, and collisions are checked with attached objects as well as between the robot's own links, so every configuration the planner tries is checked in the transfer mode's free space.
- **The planner:** OMPL's PRM or RRT plans each leg; the planner's path has no timing.

- **Timing:** an adapter after the planner turns the path into a trajectory that keeps each joint within its speed and acceleration limits.
- **Reuse:** MoveIt can keep a PRM's roadmap between requests, the multi-query (G-2786) reuse of the probabilistic roadmaps; its lazy versions check edges again only as a route needs them, by lazy collision checking (G-2799), which suits a scene that changes a little, such as a cup that has moved from A to B (MoveIt docs, "OMPL Planner", persistent roadmaps).

MoveIt's pick-and-place tutorial builds the task of Section 2 from stages: move to pick (a transit), approach and grasp, attach the object, lift, move to place (a transfer), lower and release, retreat (MoveIt docs, pick and place tutorial). Its lift and lower stages are the short final motions of Section 4.4.

6. Summary

Idea	What it does	On the cup task
Transit and transfer	split the task into motions with the hand empty and with the object held	free space 49.4, 42.3 and 51.3 percent in the three stages
Transition configuration	where one mode can hand over to the other	grasp (34.0, −144.1) degrees, placing (23.2, −67.0)
Manipulation planning	plans each leg in its mode's free space	legs of 169.0, 123.2 and 99.2 degrees; the cup clears the bottle by 0.036 m
Nonprehensile manipulation	moves objects without holding them	the open cup pushed 6 cm to C
Quasistatic assumption	forces balance at every instant of a slow push	the cup stops within 0.32 mm of the hand; 3.2 cm past at 0.5 m/s
Push height	slide or tip, from the torques about the front corner	slides below $w/(2\mu) = 0.075$ m
Preimage, compliant motion	the starts from which a command surely succeeds; sliding enlarges them	3 cm by 8.5 cm with sliding, a 1 cm sliver without
Preimage planning	chains commands backwards from the goal	drop from up to 2.8 cm above the posts
MoveIt and OMPL	arm software and its library of planners	PRM, attached cup, timing

In the table, w is the cup's width, 0.06 m, and μ the coefficient of friction, 0.4.

- The robot moves the cup by planning for the cup as well as the arm: carried from A to B it clears the bottle by 0.036 m, where a route planned for the arm alone dragged it 1.7 cm into the table and through the bottle for 32.2 percent of the way, because the transfer is planned in the free space in which the held cup is part of the robot; pushed slowly, the open cup stops within 0.32 mm of where the hand stops; and dropped from up to 2.8 cm above the holder, it lands in the holder despite the hand's 1 cm error.
- A **transit** (G-2801) and a **transfer** (G-2802) each have their own free space, because the cup is an obstacle while it stands and part of the arm while it is held; the **mode** (G-2803) says which applies.
- The mode changes only at a **transition configuration** (G-2804), free in both modes, so the grasp and the placing are where the plan's legs join.
- **Manipulation planning** (G-2800) plans each leg with an ordinary planner, here the probabilistic roadmap, in that leg's free space, so planners for fixed obstacles carry over to moving objects.
- **Nonprehensile manipulation** (G-2805) moves objects that cannot be grasped, such as the open cup, by pushing, so the robot is not limited to what its gripper can hold.

- Under the **quasistatic assumption** (G-2806) the forces balance at every instant, because speeding a slowly pushed cup up takes 2.5 percent of the 0.981 N of friction, given by the **coefficient of friction** (G-2807) and **Newton's second law** (G-2809); so the cup stops within 0.32 mm of where the hand stops, and the plan only needs the hand's path.
- Comparing **torques** (G-2808) about the cup's front corner shows where to push, because the weight's torque must exceed the push's: below 7.5 cm the cup slides, above it the cup tips.
- The **preimage** (G-2811) of the holder for "move down" is a 1 cm sliver without sliding and a triangle 3 cm wide and 8.5 cm tall with **compliant motion** (G-2810), because the chamfers make the cup slide into the slot instead of stopping; so a hand known to within 1 cm can place the cup from up to 2.8 cm above the posts.
- **Preimage planning** (G-2812) chains commands backwards from the goal, so the transfer must end inside the drop's preimage, not at an exact placing.
- **MoveIt** (G-2813) and **OMPL** (G-2814) supply these pieces on real robots: OMPL the planners, MoveIt the **planning scene** (G-2815) with attached objects, the collision checks and the timing.

The robot can now move objects by carrying, pushing and placing them, despite errors in its hand. Every plan so far is still a path, which says where the robot goes but not how fast, and assumes the robot can stop and turn at any point; real motors have speed and turning limits, which is planning with motion limits.

7. Sources

Built from

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- Lynch, K. M. and Park, F. C. (2017). *Modern Robotics: Mechanics, Planning, and Control*. Cambridge University Press. §3.4 "Wrenches" (torque), §8.1.1 (Newton's second law, eq. 8.4), §12.2 "Contact Forces and Friction" (Coulomb friction), §12.3 "Manipulation" (eq. 12.31 and 12.32, the quasistatic assumption, the meter-stick trick, Exercise 12.23 on a pushed box). Free preprint: http://modernrobotics.org (MR §3.4, §8.1.1, §12.2, §12.3)
- Hauser, K. *Robotic Systems*, University of Illinois. Ch.12 "Advanced Topics in Planning": "Manipulation planning and locomotion planning", "Suction grippers", "Multimodal motion planning" (transit and transfer modes, transition configurations, Simple Pick-and-Place), "Pre-grasp, post-grasp, pre-placement, and post-placement motions", https://motion.cs.illinois.edu/RoboticSystems/AdvancedTopicsInPlanning.html (Hauser ch.12)
- LaValle, S. M. (2006). *Planning Algorithms*. Cambridge University Press. §7.3.2 "Manipulation Planning" (admissible, stable and grasped configurations, transit and transfer paths, manipulation graph), https://lavalle.pl/planning/node331.html to node335.html; §12.5.1 "Preimage Planning" (compliant motions, sources of uncertainty, preimages, backprojections), https://lavalle.pl/planning/node634.html

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- MoveIt documentation (PickNik Robotics), “Motion Planning” concepts page, https://moveit.picknik.ai/main/doc/concepts/motion_planning.html; “The move_group node”, https://moveit.picknik.ai/main/doc/concepts/move_group.html; “OMPL Planner” (persistent and lazy roadmaps), https://moveit.picknik.ai/main/doc/examples/ompl_ “Pick and Place with MoveIt Task Constructor”, https://moveit.picknik.ai/main/doc/tutorials/pick_an (MoveIt docs)
- Kavraki Lab, *The Open Motion Planning Library* (OMPL), home page: library contents and design, https://ompl.kavrakilab.org/ (OMPL), and its licence page, https://ompl.kavrakilab.org/license.html (OMPL licence page)
- MoveIt home page, “Why MoveIt?” (BSD licence), https://moveit.ai/ (MoveIt home page)

Other references

- Hauser, K. *Robotic Systems*, ch.10 “Motion Planning in Higher Dimensions”, https://motion.cs.illinois.edu/RoboticSystems/MotionPlanningHigherDimensions.html

8. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Transit (G-2801)	A motion in which the robot moves and the object stays where it stands, so the object counts as an obstacle; for example the arm moving from home to the cup.
Transfer (G-2802)	A motion in which the robot holds the object and the object moves as if attached to its hand, so the object counts as part of the robot and must not hit anything.
Mode (manipulation) (G-2803)	Which kind of motion applies, transit or transfer, together with where the object stands or how it is held; each mode has its own free space.
Transition configuration (G-2804)	A configuration that is free in two modes, such as the hand touching the cup’s lid, where a plan can switch from transit to transfer or back.

Term	Meaning
Manipulation planning (G-2800)	Planning the motions of a robot together with the objects it moves, as a sequence of transit and transfer motions joined at transition configurations, each planned in its own free space.
Nonprehensile manipulation (G-2805)	Moving an object without grasping it, by pushing, tilting, throwing or squeezing it, so a robot can move objects its gripper cannot hold.
Coefficient of friction (G-2807)	The ratio of the largest friction force two surfaces can give to the force pressing them together, such as 0.4 for the cup on the table; it sets how hard a sliding object must be pushed.
Newton's second law (G-2809)	The net force on a body equals its mass times its acceleration; it gives the force needed to change an object's speed, which the quasistatic assumption neglects.
Quasistatic assumption (G-2806)	The assumption that motions are slow enough for the force needed to change speeds to be ignored, so the forces on an object balance at every instant; a slowly pushed object then moves only while it is pushed.
Torque (G-2808)	The turning effect of a force about a point: the force times its distance from the point, measured square to the force's line; comparing torques decides whether a pushed object tips.
Preimage (G-2811)	The set of starts from which a motion command is sure to reach its goal whatever error occurs; a command is safe to issue only from inside it.
Compliant motion (G-2810)	A motion that slides along a surface while touching it, guided by the surface, such as a cup sliding down a chamfer into its holder; it lets uncertain motions still reach a goal.
Preimage planning (G-2812)	Planning a sequence of motion commands backwards from the goal, each command chosen to end inside the preimage of the next, so the sequence succeeds despite errors in motion and sensing.

Term	Meaning
OMPL (Open Motion Planning Library) (G-2814)	A library of sampling-based planners, such as PRM and RRT, that by design has no collision checker or robot model, so it can be joined to any; MoveIt uses it by default.
MoveIt (G-2813)	Open-source motion-planning software for robot arms: it keeps a model of the robot and its surroundings, calls planners such as OMPL's, checks collisions including carried objects, and times the result.
Planning scene (MoveIt) (G-2815)	MoveIt's model of the world round the robot and of the robot's state, including objects attached to the hand, which answers whether each configuration a planner tries is free.
Probabilistic roadmap (PRM) (G-2780)	A graph of random free configurations (milestones) joined by straight edges that pass a collision check, built once and then searched with A*; it describes the free space without building the obstacle region, so it scales to robots with many joints.
Symbolic task planning (G-2600)	Choosing a sequence of actions that reaches a goal, from a description of the world in true-or-false facts such as "the cup is on the table"; it decides the order of steps (go to the table, pick up, go to the door, put down) that a route search alone cannot.
Configuration (G-2286)	The list of numbers that fixes the position of every point of a robot, such as $q = (x, y, \theta) = (2, 1, 0.524)$ for a floor robot or the joint angles (θ_1, θ_2) of a 2R arm; with it a planner treats the whole robot as one point.
Milestone (G-2781)	A free configuration kept as a vertex of a probabilistic roadmap, drawn at random; milestones are the places the roadmap's routes pass through.
Minkowski sum (G-2653)	The set of all sums of a point of one shape and a point of the other, $A \oplus B$, like sweeping one shape over the other; the obstacle region of a robot that keeps its heading is the obstacle's Minkowski sum with the robot reflected through its centre.

Term	Meaning
Narrow passage (G-2793)	A thin part of free space that a route must cross, such as a gap only slightly wider than the robot; uniform random samples rarely land in it, so roadmaps tend to split on either side.
Multi-query planner (G-2786)	A planner that invests time in a reusable description of the free space, such as a roadmap, so that many later problems in the same unchanged space are solved quickly.
Lazy collision checking (G-2799)	Delaying an edge's collision check until a candidate route uses it, deleting edges found to collide and searching again; it saves the checks of the many edges no route uses.